

CPSC 8420 Advanced Machine Learning

Gradient Descent

Motivation

Gradient Descent is one of the most widely used methods in machine learning to optimize solution.

- Most problems in machine learning they don't have a closed solution like Least Squares or PCA, thus an iterative updating algorithm may be more applicable.
- Obtaining a closed solution maybe computationally demanding with operations such as matrix inversion or SVD, *etc.*
- Gradient Descent works well not only for convex problem, but also non-convex one. Though global minimum can not be guaranteed, most of times, local minimum is acceptable.

Learning Outcomes

Our goal for today's lecture is to understand:

- Why Gradient Descent decreases the objective
- Convergence rate of GD in various conditions
- Variations of GD

Gradient Descent

Differentiable unconstrained minimization

$$\min_x f(x), \quad s.t. \quad x \in \mathbb{R}^n \quad (1)$$

where f (objective or cost function) is differentiable

Iterative descent algorithms

Start with a point x^0 , and construct a sequence $\{x^t\}$ such that:

$$f(x^{t+1}) < f(x^t), \quad t = 0, 1, 2 \dots \quad (2)$$

In each iteration, search in descent direction

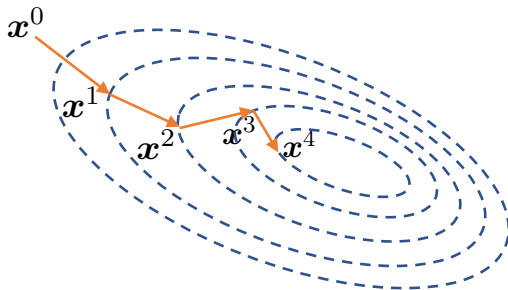
$$x^{t+1} = x^t + \eta_t d^t, \quad (3)$$

where d^t is the descent direction at x^t ; $\eta_t > 0$ is the stepsize.

Gradient descent (GD)

One of the most important examples is **Gradient descent (GD)**:

$$x^{t+1} = x^t - \eta_t \nabla f(x^t). \quad (4)$$



It can be traced to Augustin Louis Cauchy.

Gradient descent (GD)

One of the most important examples is **Gradient descent (GD)**:

$$x^{t+1} = x^t - \eta_t \nabla f(x^t). \quad (5)$$

descent direction: $d^t = -\nabla f(x^t)$, which can be justified from Taylor Expansion:

$$f(x^{t+1}) \approx f(x^t) + \langle x^{t+1} - x^t, \nabla f(x^t) \rangle + \frac{1}{2} \langle x^{t+1} - x^t, \nabla^2 f(x^t)(x^{t+1} - x^t) \rangle. \quad (6)$$

Convergence of Gradient Descent

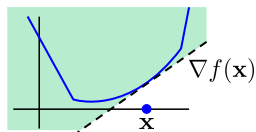
Strongly Convex Strongly Smooth Function

We say a continuously differentiable function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is α -strongly convex (SC) and β -strongly smooth (SS) if for every $x, x^+ \in \mathbb{R}^p$, we have:

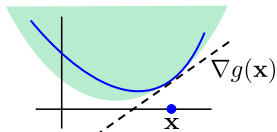
$$\frac{\alpha}{2} \|x^+ - x\|_2^2 \leq f(x^+) - f(x) - \langle \nabla f(x), x^+ - x \rangle \leq \frac{\beta}{2} \|x^+ - x\|_2^2, \quad (7)$$

α, β can be taken as the smallest and largest singular value of $\nabla^2 f(x)$.

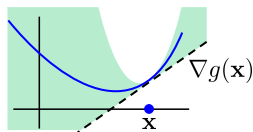
Convex, SC and SS



— $f : \mathbb{R}^d \rightarrow \mathbb{R}$
CONVEX FUNCTION

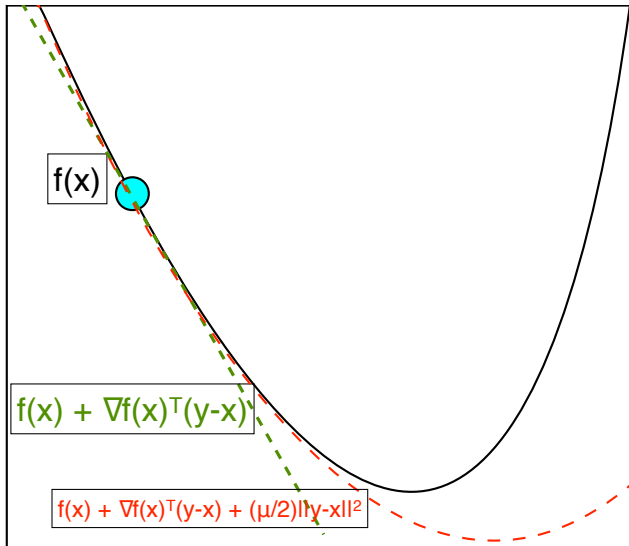


— $g : \mathbb{R}^d \rightarrow \mathbb{R}$
STRONGLY CONVEX
FUNCTION



— $g : \mathbb{R}^d \rightarrow \mathbb{R}$
STRONGLY SMOOTH
CONVEX FUNCTION

Convex, SC and SS



Strongly Convex Strongly Smooth Function

$$\frac{\alpha}{2} \|x^+ - x\|_2^2 \leq f(x^+) - f(x) - \langle \nabla f(x), x^+ - x \rangle \leq \frac{\beta}{2} \|x^+ - x\|_2^2, \quad (8)$$

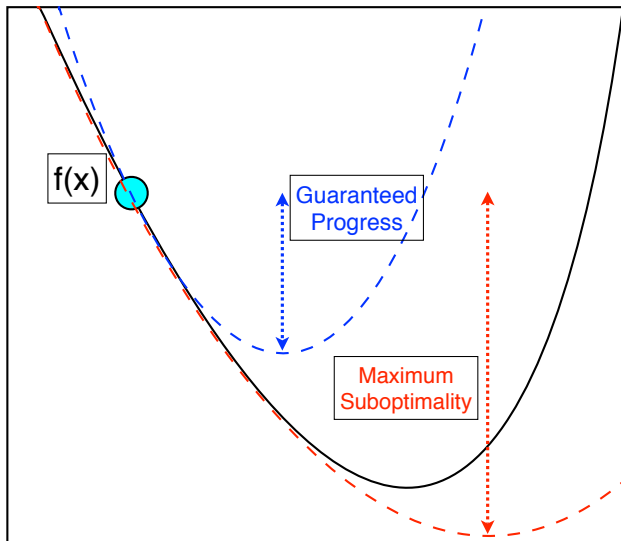
based on which we will have:

$$\begin{aligned} f(x^+) &\leq f(x) - \frac{1}{2\beta} \|\nabla f(x)\|^2 \\ f(x^+) &\geq f(x) - \frac{1}{2\alpha} \|\nabla f(x)\|^2. \end{aligned} \quad (9)$$

Replacing x^+ with x^* , then we will have:

$$\frac{1}{2\beta} \|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{1}{2\alpha} \|\nabla f(x)\|^2 \quad (10)$$

Progress within each Update



Linear Convergence Rate

$$\begin{aligned}f(x^+) - f(x^*) &\leq f(x) - f(x^*) - \frac{1}{2\beta} \|\nabla f(x)\|^2 \\&\leq f(x) - f(x^*) - \frac{\alpha}{\beta} (f(x) - f(x^*)) \\&= \left(1 - \frac{\alpha}{\beta}\right) (f(x) - f(x^*)) \\&\leq \exp^{-\frac{\alpha}{\beta}} (f(x) - f(x^*))\end{aligned}\tag{11}$$

which implies $\frac{f(x^+) - f(x^*)}{f(x) - f(x^*)} = 1 - \frac{\alpha}{\beta}$, is the definition of linear convergence. $\frac{f(x^t) - f(x^*)}{f(x^0) - f(x^*)} \leq \exp^{-\frac{\alpha}{\beta} t}$ Then to obtain ϵ -suboptimal result, we need $\mathcal{O}(\kappa \log \frac{1}{\epsilon})$ iterations, where we define $\kappa = \frac{\beta}{\alpha}$ to be the **condition number**.

Revisiting Ridge Regression

Vanilla Least Squares:

$$\min_{\beta} \frac{1}{2} \|y - A\beta\|^2 \quad (12)$$

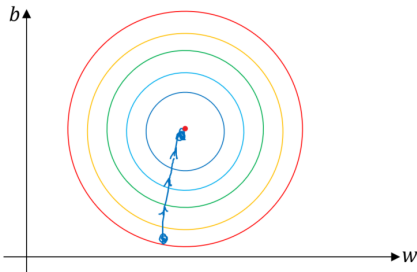
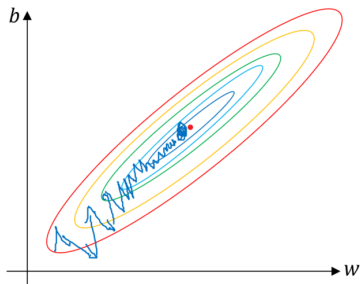
which is a convex problem with respect to β , and we can verify that $\nabla^2 f(\beta) = A^T A$, thus $\kappa = \frac{\sigma_{\max}(A^T A)}{\sigma_{\min}(A^T A)}$.

Vanilla Least Squares:

$$\min_{\beta} \frac{1}{2} \|y - A\beta\|^2 + \frac{\lambda}{2} \|\beta\|^2 \quad (13)$$

which is also convex with respect to β , and we can verify that $\nabla^2 f(\beta) = A^T A + \lambda I$, thus the new $\tilde{\kappa} = \frac{\sigma_{\max}(A^T A) + \lambda}{\sigma_{\min}(A^T A) + \lambda} < \kappa$.

Geometry of κ to influence convergence rate



Non Strongly Convex

$$\begin{aligned} f(x^+) - f(x^*) &\leq f(x) - f(x^*) - \frac{1}{2\beta} \|\nabla f(x)\|^2 \\ &\leq \langle \nabla f(x), x - x^* \rangle - \frac{1}{2\beta} \|\nabla f(x)\|^2 \end{aligned} \quad (14)$$

on the other hand we have:

$$\begin{aligned} \|x^+ - x^*\|^2 &= \|x - \eta \nabla f(x) - x^*\|^2 \\ &= \|x - x^*\|^2 - 2\eta \langle \nabla f(x), x - x^* \rangle + \eta^2 \|\nabla f(x)\|^2 \\ &= \|x - x^*\|^2 - 2\eta (\langle \nabla f(x), x - x^* \rangle - \frac{\eta}{2} \|\nabla f(x)\|^2), \end{aligned} \quad (15)$$

then we have

$\langle \nabla f(x), x - x^* \rangle - \frac{\eta}{2} \|\nabla f(x)\|^2 = \frac{1}{2\eta} (\|x - x^*\|^2 - \|x^+ - x^*\|^2)$, and
therefore $f(x^+) - f(x^*) \leq \frac{1}{2\eta} (\|x - x^*\|^2 - \|x^+ - x^*\|^2)$

Non Strongly Convex

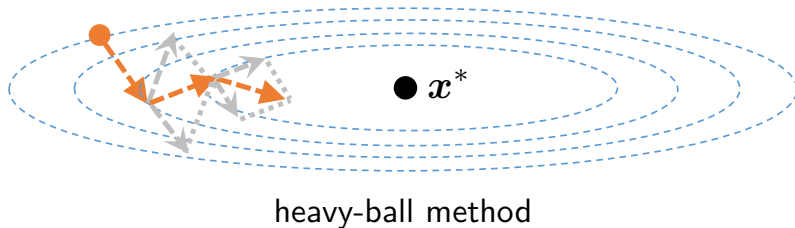
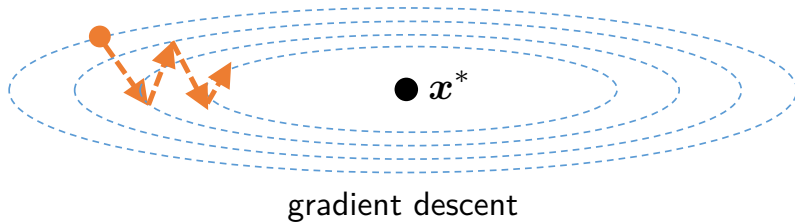
Summation the equation above from $k = 0$ to $k = T - 1$, we have:

$$\sum_{k=0}^{T-1} f(x_{k+1}) - f(x^*) \leq \frac{\|x_0 - x^*\|^2 - \|x_T - x^*\|^2}{2\eta} \leq \frac{\|x_0 - x^*\|^2}{2\eta}, \text{ then}$$

$f(x_T) - f(x^*) \leq \frac{1}{T} \sum_{k=0}^{T-1} f(x_{k+1}) - f(x^*) \leq \frac{\|x_0 - x^*\|^2}{2T\eta}$, which is sub-linear convergence rate. Then to obtain ϵ -suboptimal result, we need $T = \mathcal{O}(\frac{1}{\epsilon})$ iterations.

Variations

Heavy-ball



Heavy-ball



B. Polyak

$$\text{minimize}_{\mathbf{x} \in \mathbb{R}^n} \quad f(\mathbf{x})$$

$$\mathbf{x}^{t+1} = \mathbf{x}^t - \eta_t \nabla f(\mathbf{x}^t) + \underbrace{\theta_t (\mathbf{x}^t - \mathbf{x}^{t-1})}_{\text{momentum term}}$$

- add inertia to the “ball” (i.e. include a momentum term) to mitigate zigzagging

Heavy-ball

iteration complexity: $O(\sqrt{\kappa} \log \frac{1}{\varepsilon})$

significant improvement over GD: $O(\sqrt{\kappa} \log \frac{1}{\varepsilon})$ vs. $O(\kappa \log \frac{1}{\varepsilon})$

Nesterov Accelerate Gradient



Y. Nesterov

— Nesterov '83

$$\mathbf{x}^{t+1} = \mathbf{y}^t - \eta_t \nabla f(\mathbf{y}^t)$$

$$\mathbf{y}^{t+1} = \mathbf{x}^{t+1} + \frac{t}{t+3}(\mathbf{x}^{t+1} - \mathbf{x}^t)$$

Nesterov Accelerate Gradient

Iteration complexity: $\mathcal{O}(\frac{1}{\sqrt{\epsilon}})$, much faster than gradient descent: $\mathcal{O}(\frac{1}{\epsilon})$, alternates between gradient updates and proper extrapolation. It is not a descent method (i.e. we may not have $f(x^{t+1}) \leq f(x^t)$). It is one of the **most beautiful and mysterious** results in optimization.

Conclusion

	stepsize rule	convergence rate	iteration complexity
convex & smooth problems	$\eta_t = \frac{1}{L}$	$O\left(\frac{1}{t^2}\right)$	$O\left(\frac{1}{\sqrt{\varepsilon}}\right)$
strongly convex & smooth problems	$\eta_t = \frac{1}{L}$	$O\left(\left(1 - \frac{1}{\sqrt{\kappa}}\right)^t\right)$	$O\left(\sqrt{\kappa} \log \frac{1}{\varepsilon}\right)$